



"Life! Don't talk to me about life."

A-tractor beam

The term 'Tractor Beam' was coined by E.E. Smith, a Sci-Fi writer of some renown in the 1930s. It was a recoinning of his first name for it (the 'attractor beam'. Guess that was a little too on-the-nose). The tractor beam was made famous in pop sci-fi by Lucas and Roddenberry in their respective signature series. In reality too, the concept was theorized several times over the past 80+ years, with several fundamental theories of gravity-like beams, and was even reported to have been tested by Nikola Tesla.

There was no major breakthrough, however, until back in 2010, some scientists in ANU (Prof. Andrei Rode and his team) successfully managed to design a laser beam that could move very small particles up to distances of a metre and a half using nothing except the power of light. Earlier this year, some scientists from Singapore and Florida (Dr. Aristide Dogariu and his team) developed a beam of unstructured light that could manipulate objects over macroscopic distances along a chosen interface of two materials with different refractive indices.

It may take a little time for Tractor Beams to go from micro to macro, but as history goes to show - anything's possible.

Almost there. Almost.

There are some sci-fi-centric concepts that are brilliant in themselves, but we haven't made as much progress in them as we would like to: Teleportation and Invisibility, for example.

Teleportation has two working theories for its execution. The first, simpler one is the Trekkie way. In the simplest possible explanation: Disassemble the subject at one end, store it in data form, send it to the other end and put it back together.

Which makes perfect sense. Theoretically.

But the human body isn't a piece of hardware. Even if someone is successfully pulled apart and put together, there's no guarantee that the person's thoughts, memories and the rest of the metaphysical content will be transferred as well. I mean, really, what's the bit space occupied by the human soul, exactly?

So scratching that idea, C. H. Bennett, G. Brassard, C. Crépeau, R. Jozsa, A. Peres and W. K. Wootters proposed the theory of quantum teleportation. They made use of two postulates of quantum theory, one was Heisenberg's Uncertainty Principle, and the other, the theory of Quantum Entanglement. These concepts might be a bit abstruse for the average reader so check out <http://dgit.in/quanport> for more information.

Bearing in mind that our quantum theory is a bit rusty at the moment, here's the TLDR version of what we understood.

What these scientists are doing is creating an exact replica of you, down to the last quark or lower, and inescapably destroying the original you as a consequence of the process and Heisenberg's uncertainty principle (you really shouldn't have skipped that physics lecture). This does raise the Ship of Theseus Paradox however. Is a replica of yourself, no matter how lossless, still as good as the original? If yes, then what are you? What is your place in the universe? What is the purpose of your existence? Are you that easily replaceable? Nothing better than a drone?

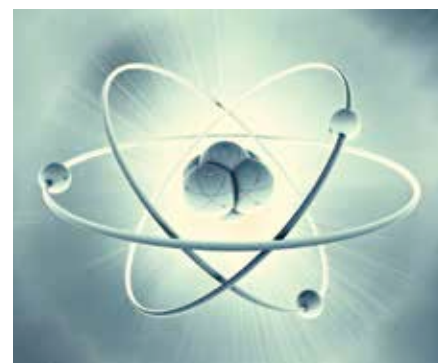
The six scientists were partially successful in teleporting a photon a distance of one metre, but with certain heavy conditions attached, making teleportation for anything other than a photon in the exact test conditions a theoretical improbability. But a few months ago, in August 2013, two teams of scientists, one in Japan and one in Switzerland made a few breakthroughs in quantum teleportation. The Japanese scientists developed a "hybrid technique involving continuous-variable-teleportation of a discrete-variable, photonic qubit." which minimizes

photon loss during the teleportation. The Swiss scientists, meanwhile, were able to teleport an electrical current between two circuits and measure the changed state without altering the current. Both experiments were breakthroughs on different levels.

So we seem to be heading in the right direction, but hold on to your molecules, because we're not quite there, yet.

Invisibility (unlike teleportation) was a much simpler concept to grasp. The theory hinged on the property of light to bend. Invisibility could be achieved by an object if it was given the ability to bend light rays around it to prevent light from reflecting off it (which is what makes an object visible). As easy as it's to say, it's equally hard to implement.

However, some time ago back in 2006, some physicists from Duke University



Quantum Psysics: The key to Teleportation

managed to bend microwaves around an object with virtually no distortion, it's effectively invisible, or cloaked, when observed at that frequency range. The invisibility isn't perfect - there's still some distortion in the microwave range - and it only works at a specific frequency of microwave radiation.

In March 2013, Andrea Alu, a physicist at the University of Texas and her colleagues successfully hid an 18 centimeter long cylindrical rod from view in microwave light. The invention was made of a new kind of material called a metascreen, created from strips of copper tape and a flexible polycarbonate film combined in a diagonal fishnet pattern.

Once again, close, but still a way to go.

Watching all that we grew up reading as sci-fi come to life around us is a reminder that Science Fiction thrives on the belief that nothing is beyond us. That we are, in fact, infinite. **d**